Rajesh Kumar, PhD

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3,900+	31	72	15	7
CITATIONS	H-INDEX	PUBLICATIONS	PATENT FILINGS	FUNDED PROJECTS

RESEARCH PROFILE

Postdoctoral researcher at Beihang University working on privacy-preserving medical AI: federated and transfer learning for medical imaging, blockchain-enabled collaboration between institutions, and security for IoT and Android systems. First-author studies address PET–CT segmentation, LLM-enabled medical imaging, secure model aggregation and Android malware detection. Current work extends this line to large language models, from LLM-enabled diagnostics to execution-grounded multi-agent frameworks.

RESEARCH INTERESTS

Medical imaging AI and computer vision; Federated and transfer learning; Privacy engineering and blockchain; Large language models; IoT cybersecurity

ACADEMIC APPOINTMENTS

2025–present
Postdoctoral Fellow
Beihang University, Hangzhou International Innovation Institute, China

2021–2025
Associate Researcher / Teaching Fellow
Yangtze Delta Region Institute (Huzhou), UESTC, China

2019–2021
Postdoctoral Fellow
University of Electronic Science and Technology of China, China
Advisor: Dr. Wang Wenyong

EDUCATION

2016–2019
Ph.D., Computer Science and Technology
University of Electronic Science and Technology of China
Thesis: Android malware detection based on machine learning
Advisor: Prof. Xiaosong Zhang

2014–2016
M.S., Computer Science
University of Sindh, Jamshoro, Pakistan

2010–2013
B.S., Computer Science
University of Sindh, Jamshoro, Pakistan

SELECTED FIRST-AUTHOR PUBLICATIONS

78 scholarly works: 47 journal articles (8 as first author), 5 other peer-reviewed articles, 19 conference papers, 1 book chapter and 6 preprints. Full list from page 3. Citations and h-index from Google Scholar, October 2026.

- **LLM-enabled medical imaging.** Integrates LLM-based medical-image analysis with a blockchain-oriented AI framework (MID-LLM). IEEE Internet of Things Journal (2025).
- **Multimodal PET–CT and medical AI.** Federated transfer learning for liver-lesion segmentation with multimodal feature integration. Artificial Intelligence in Medicine (2025).
- **Federated brain-tumor segmentation.** Privacy-preserving blockchain-based federated learning applied to brain-tumor segmentation. Computers in Biology and Medicine (2024).
- **Private model collaboration.** Blockchain and homomorphic encryption for medical-image model aggregation. Computerized Medical Imaging and Graphics (2022).
- **Distributed CT-image learning.** Federated deep-learning methods for COVID-19 CT analysis. IEEE Sensors Journal (2021); 500+ citations.
- **Secure CT-image sharing.** Blockchain-based CT-image sharing combined with AI-assisted image analysis for hospitals. Computerized Medical Imaging and Graphics (2021); 140+ citations.
- **Android and IoT security.** Multimodal malware-feature analysis and secure information-sharing. IEEE Access (2019); 150+ citations.

RESEARCH GRANTS AND FUNDED PROJECTS

2023–2026 **Intelligent image generation software based on private large language models**
PI | RMB 500,000 | Yangtze Delta Region Institute (Huzhou), UESTC | Project H01240121

2023–2025 **Blockchain-based federated learning for PET–CT imaging**
PI | RMB 50,000 | Huzhou Hospital | Project U0723003

2021–2026 **High-performance lightweight industrial blockchain technology and applications**
PI | RMB 400,000 | Yangtze Delta Region Institute (Huzhou), UESTC | Project U03210049

2021–2024 **RFID-based airline luggage tracing**
Participant | RMB 2,100,000 | National Natural Science Foundation of China | Project U2033212

2020–2021 **AI, COVID-19 recognition and blockchain data sharing**
UESTC | Project Y12519028554016524

2019–2021 **Blockchain sharding consensus**
PI | RMB 50,000 | China Postdoctoral Science Foundation | Project Y03019023601016201

2017–2019 **Android / IoT malware identification**
Participant | National Natural Science Foundation of China | Project 61572115

TEACHING EXPERIENCE

2021–2025 **University-level teaching — Yangtze Delta Region Institute (Huzhou), UESTC**
Java, database systems, data structures, C++, machine learning, deep learning, blockchain and research methodology. Delivered lectures and tutorials, developed course materials, and mentored master's research projects.

2014–2016 **Earlier instructor experience — AIMS IT Academy**
Java programming, PHP web development, Android development, database design and data structures.

Research supervision and mentoring

- Mentored master's-level research projects and thesis work.
- Organized seminars and workshops on federated learning and secure data sharing.
- Course materials on GitHub: machine learning; research writing and LaTeX; Java programming.

HONOURS AND AWARDS

- **Associate Researcher (senior professional title)**, certified by the Zhejiang Provincial Department of Human Resources and Social Security (2021).
- **Best Runner-Up**, IEEE Sensors Journal.
- **Best Oral Presentation**, International Conference on Computing and Artificial Intelligence.
- **Excellent Performance Award and Outstanding Graduate Student**, Ph.D. studies, UESTC (2016–2019).

INTELLECTUAL PROPERTY

Chinese patent applications

Blockchain-based multi-agent heterogeneous federated learning CN 2023104444528 | 2023

Blockchain-enhanced cyclic distillation-guided channel decoupling for personalized federated learning CN 2023107524401 | 2023

Autonomous navigation and obstacle avoidance system based on deep learning CN 2023104317236 | 2023

Trusted tracking system based on multi-layer blockchain CN 2023104669004 | 2023

Context-enhanced Dirichlet model for online clustering in short text streams CN 2022115045851 | 2022

RFID tag design with platform-tolerant features using multi-resonant surfaces CN 2022113882264 | 2022

Method for ranking the importance of high-order structures in high-order networks CN 2022114545198 | 2022

Deep learning framework for addressee detection using audio and facial inputs CN 2022110197167 | 2022

Potential-based method for identifying influential nodes in complex social networks CN 2022115441904 | 2022

Blockchain-based distributed demand-side scheduling for community microgrids in smart cities CN 2023112034968 | 2023

Multi-agent sensory framework for efficient decision-making with hippocampal replay and deep reinforcement learning CN 2023106365380 | 2023

Blockchain-based data management system framework CN 2023114293789 | 2023

Multi-layer response system for botnets based on complex social networks CN 2023116348381 | 2023

United Kingdom design records

Blockchain-based miniature energy meter UK 6321761 | 2023

AI-based device for detecting malware UK 6302131 | 2023

RESEARCH SOFTWARE AND DATASETS

- **Federated learning for medical images**. PyTorch federated-learning pipeline for MRI and CT images, with a BraTS preprocessing example. Code repository.
- **Blockchain-based federated learning**. Federated training in which each round's global model is recorded as a block, demonstrated on MNIST. Code repository.
- **Android malware detection**. Multi-feature detection from static features and API-call sequences, comparing deep and classical classifiers. Code repository.
- **2D / 3D image annotation materials**. Project documentation and examples for 2D/3D image annotation workflows. Code repository.
- **Camera-based robotic mower**. Demonstration materials for camera-guided robotic mower navigation. Code repository.

Datasets and research data preparation

- **Robotic grass-cutting dataset**. Image dataset for camera-guided robotic lawn-mower research.
- **Autonomous ship navigation dataset**. Six open-source datasets integrated for experiments.
- **Sindhi named-entity recognition dataset**. Linguistic data resources for Sindhi NER research.
- **COVID-19 chest CT imaging data**. Multi-hospital CT data collected and curated for detection research.

ACADEMIC SERVICE AND PEER REVIEW

- **Journal reviewer**. IEEE Transactions on Information Forensics and Security; IEEE Internet of Things Journal; Computers & Security; IEEE Access.
- **Conference and workshop reviewer**. 14th and 15th International Computer Conference on Wavelet Active Media Technology and Information Processing; Workshop on Cloud Security and Privacy.

SELECTED INVITED TALKS AND PRESENTATIONS

- **Artificial intelligence and the environment**. Keynote to young scholars, Huzhou (2021).
- **Challenges and implications of blockchain with federated learning**. Keynote, Guangdong (2019).
- **Malicious code detection using image processing and deep learning**. International Conference on Computing and Artificial Intelligence, Chengdu (2018).
- **Artificial intelligence challenges and contemporary trends**. UESTC AI seminar (2019).
- **LaTeX and research tools**. Invited talk, Sichuan University.

JOURNAL ARTICLES (47)

1. Mehdi Sedighi, **Rajesh Kumar**, Mostafa Ghasemi. ANN-guided multi-objective optimization of SPEEK/MXene nanocomposite membranes for microbial fuel cell wastewater treatment and energy recovery. *Biochemical Engineering Journal*, 237, 110407, 2027. DOI: 10.1016/j.bej.2026.110407
2. Riaz Ullah Khan, **Rajesh Kumar**, Abu Bakar Sharif. A secure, transparent, and decentralised framework for cross-domain healthcare data sharing using blockchain-based EHRs. *Journal of Cyber Security Technology*, 10(1), article 2706567, 2026. DOI: 10.1080/23742917.2026.2706567
3. Kimia Rostami, **Rajesh Kumar**, Abdalla M. Abdalla, Mehdi Sedighi, Mostafa Ghasemi. Random Forest and differential evolution for maximizing microbial fuel cell performance in wastewater remediation. *Biomass and Bioenergy*, 208, 108868, 2026. DOI: 10.1016/j.biombioe.2025.108868
4. Judith Ayekai Browne, Chen Wenyu, Khalid Odartey Lamptey, Delanyo Kwame Bensah, Edward Mensah Acheampong, **Rajesh Kumar**, Najma Imtiaz Ali, Nurul A. Emran. RePFL: lag-resilient personalized federated learning for skin cancer detection under extreme non-IID heterogeneity. *Scientific Reports*, 2026. DOI: 10.1038/s41598-026-64096-5
5. Riaz Ullah Khan, Hanan Aljuaid, Dawar Khan, **Rajesh Kumar**. BotWard: A resilient framework for detecting and mitigating botnets in complex social networks through pseudo-random nickname identification. *Peer-to-Peer Networking and Applications*, 18(6), article 303, 2025. DOI: 10.1007/s12083-025-02110-9

6. **Rajesh Kumar**, Yunbo Rao, Jay Kumar, Cobbinah B. Mawuli, Waqar Ali, Shaoning Zeng. MID-LLM: Enhancing Medical Image Diagnostics With LLMs in a Blockchain AI Framework. *IEEE Internet of Things Journal*, 12(18), 37160–37174, 2025. DOI: 10.1109/JIOT.2025.3583408
7. **Rajesh Kumar**, Shaoning Zeng, Jay Kumar, Zakria, Xinfeng Mao. Privacy-preserving federated transfer learning for enhanced liver lesion segmentation in PET-CT imaging. *Artificial Intelligence in Medicine*, 169, 103245, 2025. DOI: 10.1016/j.artmed.2025.103245
8. Ali Shahzad, Wenyu Chen, Yin Zhang, **Rajesh Kumar**. Zero-Trust Medical Image Sharing: A Secure and Decentralized Approach Using Blockchain and the IPFS. *Symmetry*, 17(4), 551, 2025. DOI: 10.3390/sym17040551
9. Abdullah Aman Khan, Khwaja Mutahir Ahmad, Sidra Shafiq, Waqas Amin, **Rajesh Kumar**. 3DFFL: Privacy-Preserving Federated Few-Shot Learning for 3D Point Clouds in Autonomous Vehicles. *Scientific Reports*, 14, 2024. DOI: 10.1038/s41598-024-70326-5
10. Yanmei Li, Jinghong Tang, Weiwu Ding, Jian Luo, Naveed Ahmad, **Rajesh Kumar**. An Assisted Diagnosis of Alzheimer's Disease Incorporating Attention Mechanisms Med-3D Transfer Modeling. *Computers, Materials & Continua*, 78(1), 713–733, 2024. DOI: 10.32604/cmc.2023.046872
11. Riaz Ullah Khan, Sultan Almakdi, Mohammed Alshehri, Amin Ul Haq, Aman Ullah, **Rajesh Kumar**. An Intelligent Neural Network Model to Detect Red Blood Cells for Various Blood Structure Classification in Microscopic Medical Images. *Heliyon*, 10(4), e26149, 2024. DOI: 10.1016/j.heliyon.2024.e26149
12. Hub Ali, Gang Xiong, Qu Tianci, **Rajesh Kumar**, Xisong Dong, Zhen Shen. Autonomous Ship Navigation with an Enhanced Safety Collision Avoidance Technique. *ISA Transactions*, 144, 271–281, 2024. DOI: 10.1016/j.isatra.2023.10.019
13. Riaz Ullah Khan, **Rajesh Kumar**, Amin Ul Haq, Inayat Khan, Mohammad Shabaz, Faheem Khan. Blockchain-Based Trusted Tracking Smart Sensing Network to Prevent the Spread of Infectious Diseases. *IRBM*, 2024. ScienceDirect PII: S1959031824000101
14. Delanyo Kwame Bensah Kulevome, Hong Wang, Bernard Mawuli Cobbinah, Ernest Smith Mawuli, **Rajesh Kumar**. Effective Time-Series Data Augmentation with Analytic Wavelets for Bearing Fault Diagnosis. *Expert Systems with Applications*, 249, 123536, 2024. DOI: 10.1016/j.eswa.2024.123536
15. **Rajesh Kumar**, Cobbinah M. Bernard, Aman Ullah, Riaz Ullah Khan, Jay Kumar, Delanyo K. B. Kulevome, Rao Yunbo, Shaoning Zeng. Privacy-preserving blockchain-based federated learning for brain tumor segmentation. *Computers in Biology and Medicine*, 177, 108646, 2024. DOI: 10.1016/j.compbiomed.2024.108646
16. Waqar Ali, **Rajesh Kumar**, Xiangmin Zhou, Jie Shao. Responsible Recommendation Services with Blockchain Empowered Asynchronous Federated Learning. *ACM Transactions on Intelligent Systems and Technology*, 15(4), Article 78, 2024. DOI: 10.1145/3633520
17. Jay Kumar, Junming Shao, **Rajesh Kumar**, Salah Ud Din, Cobbinah B. Mawuli, Qinli Yang. A Context-Enhanced Dirichlet Model for Online Clustering in Short Text Streams. *Expert Systems with Applications*, 228, 120262, 2023. DOI: 10.1016/j.eswa.2023.120262
18. Wei Jiang, Qiulin Qian, Wangwen Yong, **Rajesh Kumar**, Junrui Wu, Hanwen Zhang. Application of No-Light Fault Prediction of PON Based on Deep Learning Method. *Computer Communications*, 208, 210–219, 2023. DOI: 10.1016/j.comcom.2023.06.002
19. Abubakar Sharif, **Rajesh Kumar**, Turke Althobaiti, Abdullah Alhumaidi Alotaibi, Lubna Safi, Naeem Ramzan, Muhammad Ali Imran, Qammer H. Abbasi. Bio-Inspired Circular Polarized UHF RFID Tag Design Using Characteristic Mode Analysis. *IEEE Sensors Journal*, 2023. IEEE Xplore: 10090430
20. Aman Ullah, Junming Shao, Qinli Yang, Nasrullah Khan, Cobbinah M. Bernard, **Rajesh Kumar**. LSS: A Locality-Based Structure System to Evaluate the Spreader's Importance in Social Complex Networks. *Expert Systems with Applications*, 228, 120326, 2023. DOI: 10.1016/j.eswa.2023.120326
21. Amin Ul Haq, Jian Ping Li, **Rajesh Kumar**, Zafar Ali, Inayat Khan, M. Irfan Uddin, Bless Lord Y. Agbley. MCNN: A Multi-Level CNN Model for the Classification of Brain Tumors in IoT-Healthcare System. *Journal of Ambient Intelligence and Humanized Computing*, 14(5), 4695–4706, 2023. DOI: 10.1007/s12652-022-04373-z
22. Abubakar Sharif, **Rajesh Kumar**, Kamran Arshad, Khaled Assaleh, Hassan Tariq Chattha, Muhammad Ali Imran, Qammer H. Abbasi. Nature-Inspired Spider Web Shaped UHF RFID Reader Antenna for IoT and Healthcare. *Scientific Reports*, 13, 2023. DOI: 10.1038/s41598-023-39825-9
23. Jay Kumar, Junming Shao, **Rajesh Kumar**, Salah Ud Din, Cobbinah B. Mawuli, Qinli Yang. Online Semi-Supervised Classification on Multi-Label Evolving High-Dimensional Text Streams. *IEEE Transactions on Systems, Man, and Cybernetics*, 2023. IEEE Xplore: 10143404
24. Delanyo Kwame Bensah Kulevome, Hong Wang, Xuegang Wang, **Rajesh Kumar**, Bernard Cobbinah. Phased Array Antenna Diagnosis from Amplitude-Only Data Using Parallel Deep Learning Models. *Journal of Applied Remote Sensing*, 17(1), 017502, 2023. DOI: 10.1117/1.JRS.17.017502
25. Abdus Saboor, Arif Hussain, Bless Lord Y. Agbley, Amin ul Haq, Jian Ping Li, **Rajesh Kumar**. Stock Market Index Prediction Using Machine Learning and Deep Learning Techniques. *Intelligent Automation & Soft Computing*, 37(2), 2023. techscience.com/iasc/v37n2/53230

26. Saifullah Tumrani, Wazir Ali, **Rajesh Kumar**, Abdullah Aman Khan, Fayaz Ali Dharejo. View-Aware Attribute-Guided Network for Vehicle Re-Identification. *Multimedia Systems*, 2023. DOI: 10.1007/s00530-023-01077-y
27. Fiseha B. Tesema, Jason Gu, Wei Song, Hong Wu, Shiqiang Zhu, Zheyuan Lin, Min Huang, Wen Wang, **Rajesh Kumar**. Addressee Detection Using Facial and Audio Features in Mixed Human-Human and Human-Robot Settings: A Deep Learning Framework. *IEEE Systems, Man, and Cybernetics Magazine*, 2022. IEEE Xplore: 10104108
28. Jay Kumar, Salah Ud Din, Qinli Yang, **Rajesh Kumar**, Junming Shao. An Online Semantic-Enhanced Graphical Model for Evolving Short Text Stream Clustering. *IEEE Transactions on Cybernetics*, 52(12), 13809–13820, 2022. DOI: 10.1109/TCYB.2021.3108897
29. **Rajesh Kumar**, Jay Kumar, Abdullah Aman Khan, Zakria, Hub Ali, Cobbinah M. Bernard, Riaz Ullah Khan, Shaoning Zeng. Blockchain and homomorphic encryption based privacy-preserving model aggregation for medical images. *Computerized Medical Imaging and Graphics*, 102, 102139, 2022. DOI: 10.1016/j.compmedimag.2022.102139
30. Zakria, Jianhua Deng, **Rajesh Kumar**, Muhammad Saddam Khokhar, Jingye Cai, Jay Kumar. Multi Scale and Direction Target Detecting in Remote Sensing Images via Modified YOLO-v4. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2022. IEEE Xplore: 9672697
31. Riaz Ullah Khan, Sultan Almakdi, Mohammed Alshehri, **Rajesh Kumar**, Ikram Ali, Sardar Muhammad Hussain, Amin Ul Haq, Inayat Khan, Aman Ullah, Muhammad Irfan Uddin. Probabilistic Approach to COVID-19 Data Analysis and Forecasting Future Outbreaks Using a Multi-Layer Perceptron Neural Network. *Diagnostics*, 12(10), 2539, 2022. DOI: 10.3390/diagnostics12102539
32. Muhammad Husnain Haider, Zhonglai Wang, Abdullah Aman Khan, Hub Ali, Hao Zheng, Shaban Usman, **Rajesh Kumar**, M. Usman Maqbool Bhutta, Pengpeng Zhi. Robust Mobile Robot Navigation in Cluttered Environments Based on Hybrid Adaptive Neuro-Fuzzy Inference and Sensor Fusion. *Journal of King Saud University - Computer and Information Sciences*, 2022. ScienceDirect PII: S1319157822003135
33. Waqar Ali, **Rajesh Kumar**, Zhiyi Deng, Yansong Wang, Jie Shao. A Federated Learning Approach for Privacy Protection in Context-Aware Recommender Systems. *The Computer Journal*, 64(7), 2021. academic.oup.com/comjnl/article/64/7/1016/6259634
34. Ikram Ali, Yong Chen, Niamat Ullah, **Rajesh Kumar**, Wen He. An Efficient and Provably Secure ECC-Based Conditional Privacy-Preserving Authentication for Vehicle-to-Vehicle Communication in VANETs. *IEEE Transactions on Vehicular Technology*, 2021. IEEE Xplore: 9321754
35. **Rajesh Kumar**, WenYong Wang, Jay Kumar, Ting Yang, Abdullah Khan, Wazir Ali, Ikram Ali. An Integration of blockchain and AI for secure data sharing and detection of CT images for the hospitals. *Computerized Medical Imaging and Graphics*, 87, 101812, 2021. DOI: 10.1016/j.compmedimag.2020.101812
36. **Rajesh Kumar**, Abdullah Aman Khan, Jay Kumar, Zakria, Noorbakhsh Amiri Golilarz, Simin Zhang, Yang Ting, Chengyu Zheng, Wenyong Wang. Blockchain-Federated-Learning and Deep Learning Models for COVID-19 Detection Using CT Imaging. *IEEE Sensors Journal*, 21(14), 16301–16314, 2021. DOI: 10.1109/JSEN.2021.3076767
37. Wazir Ali, Jay Kumar, Zenglin Xu, **Rajesh Kumar**, Yazhou Ren. Context-Aware Bidirectional Neural Model for Sindhi Named Entity Recognition. *Applied Sciences*, 11(19), 9038, 2021. DOI: 10.3390/app11199038
38. Abubakar Sharif, **Rajesh Kumar**, Jun Ouyang, Hasan T. Abbas, Akram Alomainy, Kamran Arshad, Khaled Assaleh, Ayman Althuwayb, Muhammad Ali Imran, Qammer H. Abbasi. Making Assembly Line in Supply Chain Robust and Secure Using UHF RFID. *Scientific Reports*, 11, 2021. DOI: 10.1038/s41598-021-97598-5
39. Yousif Abuidris, **Rajesh Kumar**, Ting Yang, Joseph Onginjo. Secure Large-Scale E-Voting System Based on Blockchain Contract Using a Hybrid Consensus Model Combined with Sharding. *ETRI Journal*, 43(2), 357–370, 2021. DOI: 10.4218/etrij.2019-0362
40. Zakria, Jianhua Deng, Yang Hao, Muhammad Saddam Khokhar, **Rajesh Kumar**, Jingye Cai, Jay Kumar, Muhammad Umar Aftab. Trends in Vehicle Re-Identification Past, Present, and Future: A Comprehensive Review. *Mathematics*, 9(24), 3162, 2021. DOI: 10.3390/math9243162
41. Noorbakhsh Amiri Golilarz, Hui Gao, **Rajesh Kumar**, Liaqat Ali, Yan Fu, Chun Li. Adaptive Wavelet Based MRI Brain Image De-Noising. *Frontiers in Neuroscience*, 2020. PMCID: PMC7388743
42. Abida Sharif, Jian Ping Li, Muhammad Asim Saleem, Tanzila Saba, **Rajesh Kumar**. Efficient Hybrid Clustering Scheme for Data Delivery Using Internet of Things Enabled Vehicular Ad Hoc Networks in Smart City Traffic Congestion. *Journal of Internet Technology*, 2020. jit.ndhu.edu.tw/article/view/2231
43. Zakria, Jianhua Deng, Jingye Cai, Muhammad Umar Aftab, Muhammad Saddam Khokhar, **Rajesh Kumar**. Visual Features with Spatio-Temporal-Based Fusion Model for Cross-Dataset Vehicle Re-Identification. *Electronics*, 9(7), 1083, 2020. DOI: 10.3390/electronics9071083
44. **Rajesh Kumar**, Xiaosong Zhang, Wenyong Wang, Riaz Ullah Khan, Jay Kumar, Abubaker Sharif. A Multimodal Malware Detection Technique for Android IoT Devices Using Various Features. *IEEE Access*, 7, 64411–64430, 2019. DOI: 10.1109/ACCESS.2019.2916886

45. Riaz Ullah Khan, Xiaosong Zhang, **Rajesh Kumar**, Abubakar Sharif, Noorbakhsh Amiri Golilarz, Mamoun Alazab. An Adaptive Multi-Layer Botnet Detection Technique Using Machine Learning Classifiers. *Applied Sciences*, 9(11), 2375, 2019. DOI: 10.3390/app9112375
46. Zakria, Jingye Cai, Jianhua Deng, Muhammad Umar Aftab, Muhammad Saddam Khokhar, **Rajesh Kumar**. Efficient and Deep Vehicle Re-Identification Using Multi-Level Feature Extraction. *Applied Sciences*, 9(7), 1291, 2019. DOI: 10.3390/app9071291
47. **Rajesh Kumar**, Xiaosong Zhang, Riaz Ullah Khan, Abubakar Sharif. Research on Data Mining of Permission-Induced Risk for Android IoT Devices. *Applied Sciences*, 9(2), 277, 2019. DOI: 10.3390/app9020277

OTHER PEER-REVIEWED ARTICLES (5)

48. Riaz Ullah Khan, Xiaosong Zhang, **Rajesh Kumar**. Analysis of ResNet and GoogLeNet Models for Malware Detection. *Journal of Computer Virology and Hacking Techniques*, 15(1), 29–37, 2019. DOI: 10.1007/s11416-018-0324-z
49. Emelia Opoku Aboagye, **Rajesh Kumar**. Simple and Efficient Computational Intelligence Strategies for Effective Collaborative Decisions. *Future Internet*, 11(1), 24, 2019. DOI: 10.3390/fi11010024
50. Emelia Opoku Aboagye, Gee C. James, **Rajesh Kumar**. Evaluating the Performance of Deep Neural Networks for Health Decision Making. *Procedia Computer Science*, 131, 866–872, 2018. DOI: 10.1016/j.procs.2018.04.288
51. Riaz Ullah Khan, **Rajesh Kumar**, Nawsher Khan, Xiaosong Zhang, Ijaz Ahad. Optimizing a Deep Learning Model in Order to Have a Robust Neural Network Topology. *International Journal of Modeling and Optimization*, 8, 2018. ijmo.org/vol8/639-K0029.pdf
52. **R. K. Tharwani**, H. N. Nizamani, Q. U. A. Nizamani, Y. Khatari, F. H. Chandio, M. S. Abbasi. Modelling Choice Selection System of a Public Sector General University in Pakistan. *Sindh University Research Journal*, 2016. sujo-old.usindh.edu.pk/index.php/SURJ/article/view/2741

CONFERENCE PROCEEDINGS (19)

53. Ali Raza, Y. Chen, Z. Gong, Y. Ding, **Rajesh Kumar**. An Ultra-Wideband Tapered Slot Antenna for Medical Microwave Imaging Applications. *Progress In Electromagnetics Research Symposium*, 2024. IEEE Xplore: 10618710
54. **Rajesh Kumar**, Y. Chen, Z. Gong, Z. Al-Huda, Ali Raza. Next-Gen Medical Collaboration Integrating Blockchain for Image Sharing. *Progress In Electromagnetics Research Symposium*, 2024. IEEE Xplore: 10617922
55. Haisheng Yu, **Rajesh Kumar**, Wenyong Wang, Ji Zhang, Zhifeng Liu, Sai Zou, Jiangchuan Yang, Leong Io Hon. Intelligent Collaborative Control of Multi-Source Heterogeneous Data Streams for Low-Power IoT: A Flow Machine Learning Approach. *International Conference on Algorithms and Architectures for Parallel Processing*, 2023. DOI: 10.1007/978-981-97-0859-8_22
56. Riaz Ullah Khan, Amin Ul Haq, Sardar Muhammad Hussain, Shakir Ullah, Sultan Almakdi, **Rajesh Kumar**, Hasrat Hussain Shah, Jianping Li. Analyzing and Battling the Emerging Variants of COVID-19 Using Artificial Neural Network and Blockchain. *International Computer Conference on Wavelet Active Media Technology and Information Processing*, 2021. IEEE Xplore: 9674142
57. Ali Shahzad, Chen Wenyu, **Rajesh Kumar**. Blockchain Based Monitoring on Trustless Supply Chain Processes. *International Conference on Cloud Computing and Big Data Analytics*, 2021. IEEE Xplore: 9442512
58. Wazir Ali, **Rajesh Kumar**, Yong Dai, Jay Kumar, Saifullah Tumrani. Neural Joint Model for Part-of-Speech Tagging and Entity Extraction. *International Conference on Machine Learning and Computing*, 2021. DOI: 10.1145/3457682.3457718
59. Jay Kumar, **Rajesh Kumar**, Amin Ul Haq, Sidra Shafiq. A Non-Parametric Multi-Lingual Clustering Model for Temporal Short Text. *International Computer Conference on Wavelet Active Media Technology and Information Processing*, 2020. IEEE Xplore: 9317342
60. Huaiyu Wen, Sufang Li, Amin Ul Haq, Jian Ping Li, **Rajesh Kumar**, Abdus Saboor, Tao Jiang. Elimination of Irrelevant Features and Heart Disease Recognition by Employing Machine Learning Algorithms Using Clinical Data. *International Computer Conference on Wavelet Active Media Technology and Information Processing*, 2020. IEEE Xplore: 9317351
61. Abdullah Aman Khan, Sidra Shafiq, **Rajesh Kumar**, Jay Kumar, Amin Ul Haq. H3DNN: 3D Deep Learning Based Detection of COVID-19 Virus Using Lungs Computed Tomography. *International Computer Conference on Wavelet Active Media Technology and Information Processing*, 2020. IEEE Xplore: 9317357
62. Riaz Ullah Khan, **Rajesh Kumar**, Mamoun Alazab, Xiaosong Zhang. A Hybrid Technique to Detect Botnets, Based on P2P Traffic Similarity. *Cybersecurity and Cyberforensics Conference*, 2019. IEEE Xplore: 8854561
63. Yousif Abuidris, **Rajesh Kumar**, Wang Wenyong. A Survey of Blockchain Based on E-Voting Systems. *Proceedings of the 2019 2nd International Conference on Blockchain Technology and Applications*, 99–104, 2019. DOI: 10.1145/3376044.3376060

64. Riaz Ullah Khan, Xiaosong Zhang, Mamoun Alazab, **Rajesh Kumar**. An Improved Convolutional Neural Network Model for Intrusion Detection in Networks. *Cybersecurity and Cyberforensics Conference*, 2019. IEEE Xplore: 8854549
65. **Rajesh Kumar**, Xiaosong Zhang, Riaz Ullah Khan, Jay Kumar, Ijaz Ahad. Effective and Explainable Detection of Android Malware Based on Machine Learning Algorithms. *International Conference on Computing and Artificial Intelligence (ICCAI)*, 35–40, 2018. DOI: 10.1145/3194452.3194465
66. Riaz Ullah Khan, Xiaosong Zhang, **Rajesh Kumar**, Emelia Opoku Aboagye. Evaluating the Performance of ResNet Model Based on Image Recognition. *International Conference on Computing and Artificial Intelligence*, 2018. DOI: 10.1145/3194452.3194461
67. **Rajesh Kumar**, Xiaosong Zhang, Riaz Ullah Khan, Ijaz Ahad, Jay Kumar. Malicious Code Detection Based on Image Processing Using Deep Learning. *International Conference on Computing and Artificial Intelligence (ICCAI)*, 81–85, 2018. DOI: 10.1145/3194452.3194459
68. Emelia Opoku Aboagye, Gee C. James, Gao Jianbin, **Rajesh Kumar**, Riaz Ullah Khan. Probabilistic Time Context Framework for Big Data Collaborative Recommendation. *International Conference on Computing and Artificial Intelligence*, 2018. DOI: 10.1145/3194452.3194458
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